



Universidade do Minho

Escola de Engenharia

Departamento de Informática

Machine Learning Introduction

MESTRADO EM ENGENHARIA DE TELECOMUNICAÇÕES E INFORMÁTICA
MESTRADO integrado EM ENGENHARIA DE TELECOMUNICAÇÕES E
INFORMÁTICA

Inteligência Artificial para as Telecomunicações

2025/2026

Decisions!!!

- The action or process of deciding something or of resolving a question (quickly and decisively).
- To make (good) decisions, we must be able to predict the future, accurately perceive the present, have insight into the will of others, and the capacity to deal with uncertainty.
- The problem of (to much) intuition!
- Although intuition can be a helpful tool, it would be a mistake to make all decisions based on a feeling.
- Intuition can provide a spark that starts something new, it's through data (knowledge) that you verify, understand, and quantify...



Image source: [22https://www.advantageperformance.com/framing-a-decision/](https://www.advantageperformance.com/framing-a-decision/)

- Contextualization
- Dimensions of Learning
- Symbolic Learning
- Classical Learning: Supervised vs unsupervised



Artificial Intelligence Contextualization

- The AI approach to knowledge representation is exposed (normally) through two paradigms:
 - Symbolic
 - It is based on logic to represent knowledge;
 - It bases the reasoning in the construction of inference systems;
 - Non-symbolic, or connectionist
 - It bases the functioning of the system on the ability to learn, generalizing;
 - Solves problems based on past knowledge or data on solving other problems;



Artificial Intelligence Contextualization

- Advantages of the symbolic approach:
 - The problem of knowledge representation is simple;
 - Great expressive power of representation languages;
 - Solidified inference mechanisms.



- Advantages of the non-symbolic approach:

- Ability to learn to solve problems;
- Ability to generalize problem solving;
- Fault-tolerant.

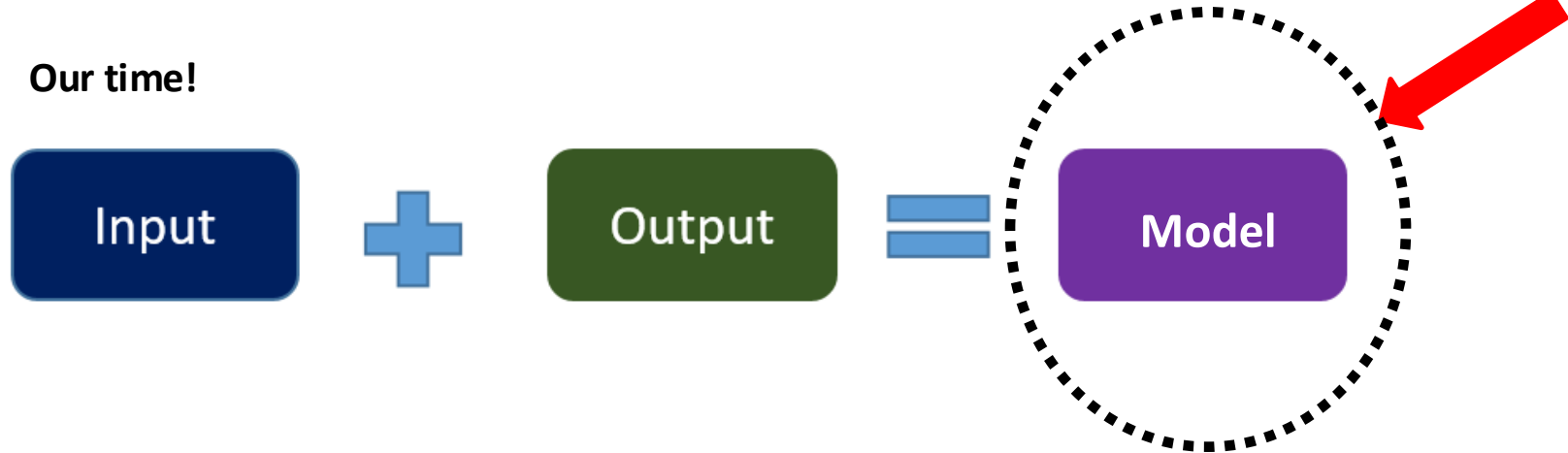


New times!

“Old times”



Our time!



Dimensions of Learning Systems

■ Type of feedback

- supervised (labeled examples)
- unsupervised (unlabeled examples)
- reinforcement (reward)

■ Representation

- attribute-based (feature vector)
- relational (first-order logic)

■ Knowledge

- empirical (knowledge-free)
- analytical (knowledge-guided)

Knowledge-based Systems: Acquisition and modeling of common-sense knowledge and expert knowledge

- ⇒ limited to given knowledge base and rule set
- ⇒ Inference: **Deduction** generates no new knowledge but makes implicitly given knowledge explicit
- ⇒ **Top-Down**: from rules to facts

Learning Systems: Extraction of knowledge and rules from examples/experience

- Teach the system vs. program the system
- Learning as **inductive process**
- ⇒ **Bottom-Up**: from facts to rules

Deduction vs Induction

Deduction	Induction
All humans are mortal. (Axiom)	Socrates is human. (Background K.)
Socrates is human. (Fact)	Socrates is mortal. (Observation(s))
<i>Conclusion:</i> Socrates is mortal.	<i>Generalization:</i> All humans are mortal.

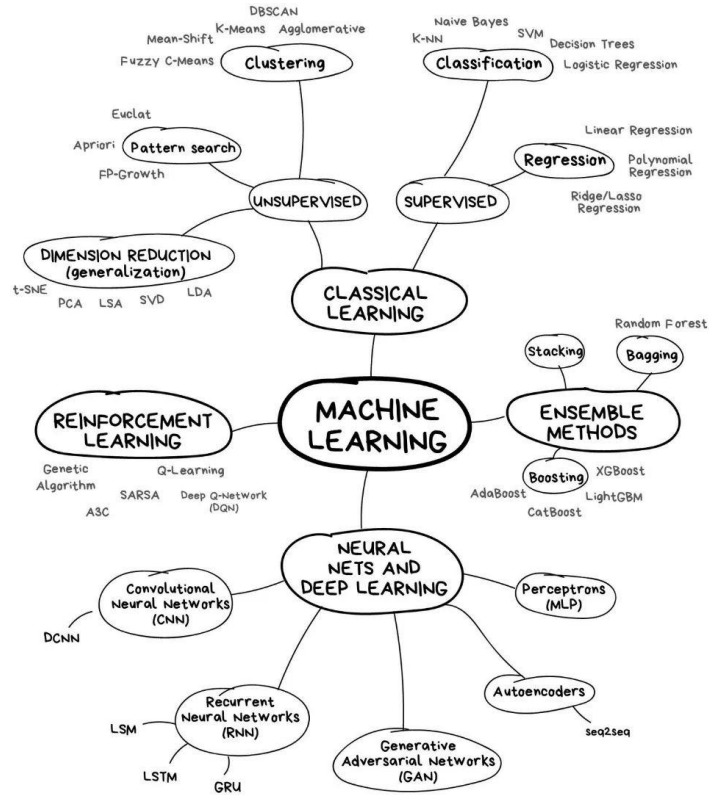
Deduction: from general to specific $\Rightarrow$ **proven** correctness

Induction: from specific to general $\Rightarrow$ **(unproven)** knowledge gain

**Induction generates hypotheses
not knowledge!**

(See also **Instance Based Learning**)

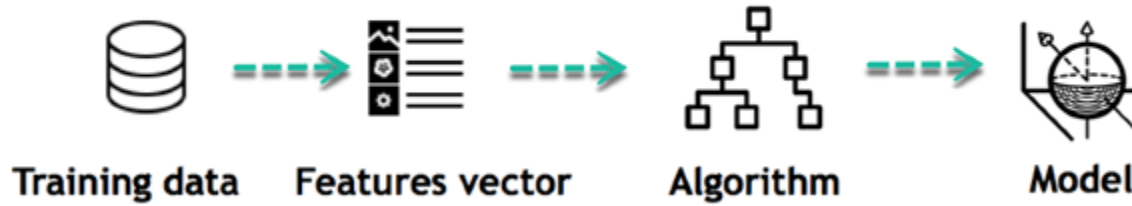
Machine Learning



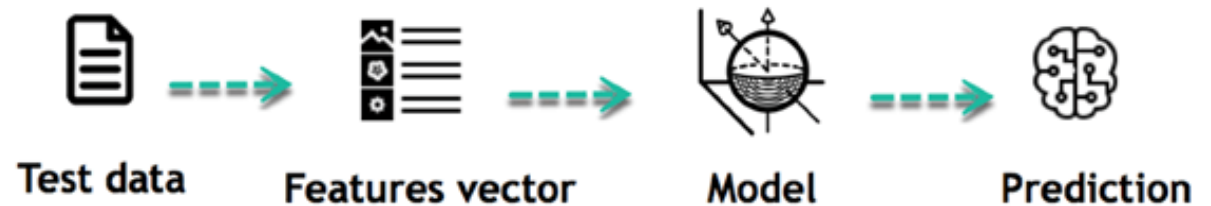
Source: The map of the machine learning world
Vasily Zubarev (vas3k.com)

How does it work?

Learning Phase



Inference from Model



- How does it happen? A simple example ...

- **Training set**

- Dataset describing a particular problem

- **Test Set**

- Data set against which the model will be tested

- **Input Variables**

- Set of variables that characterize each instance of the problem

- **Output Variables**

- Set of variables that answer the problem

Example

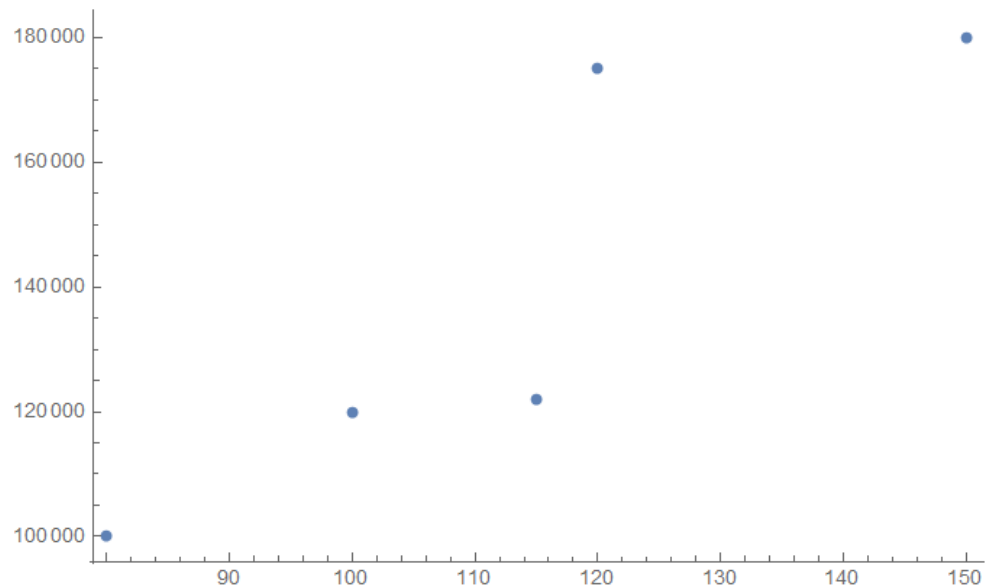
House prices by area

Area	Price
120	175000
150	180000
80	100000
100	120000
115	122000

$$m = 5$$

$$x^{(1)} = 120$$

$$y^{(3)} = 100000$$



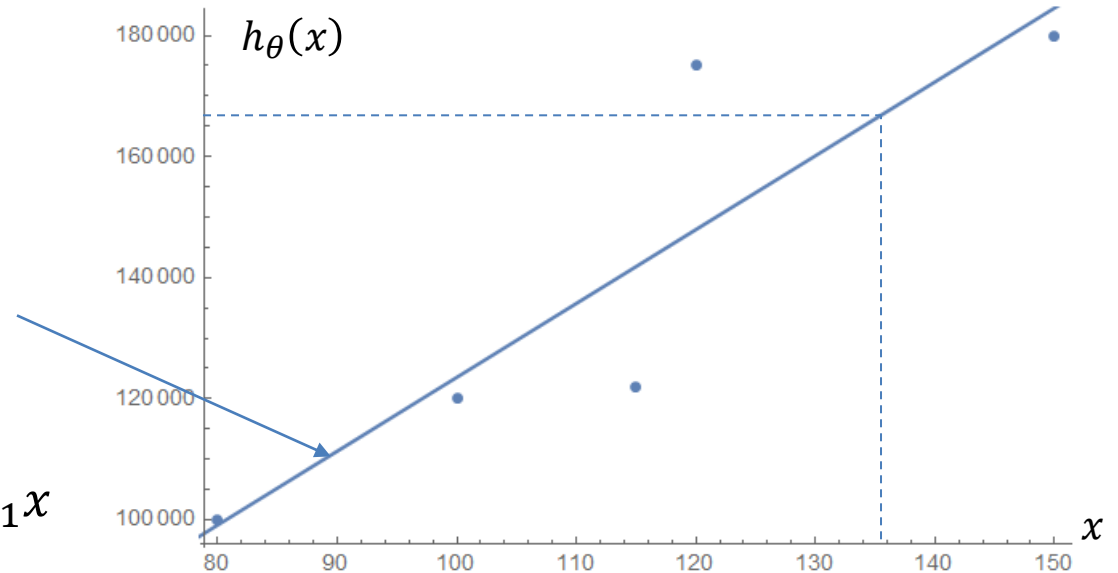
Example

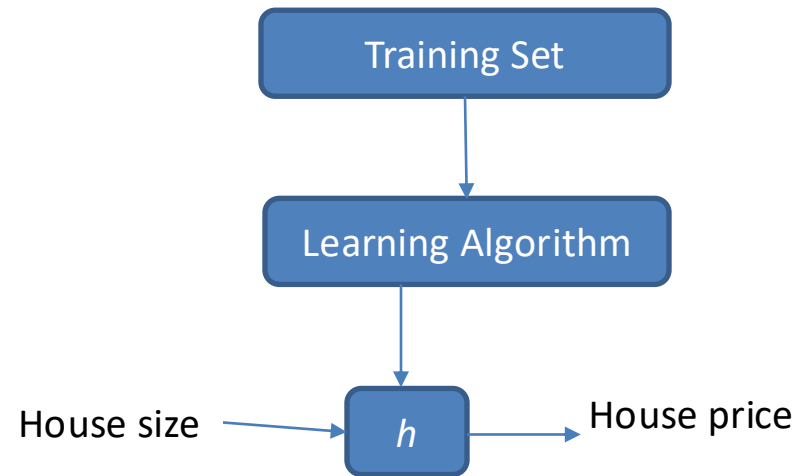
- House prices by area

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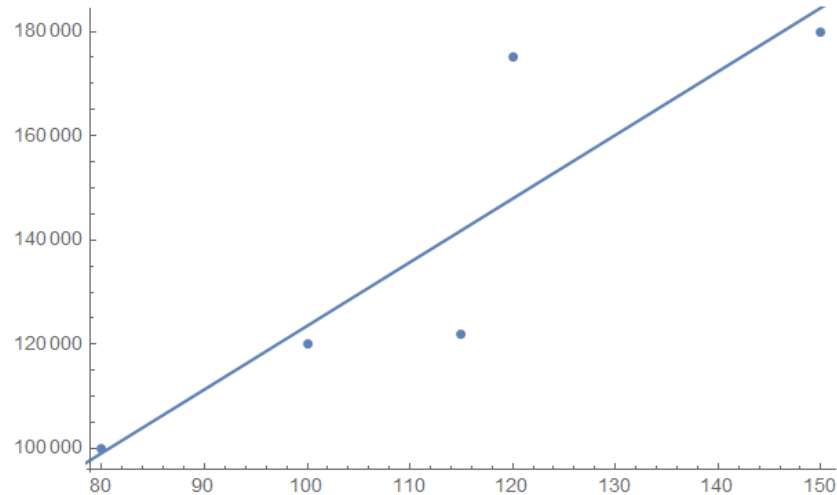
Function that
models the
problem h

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$



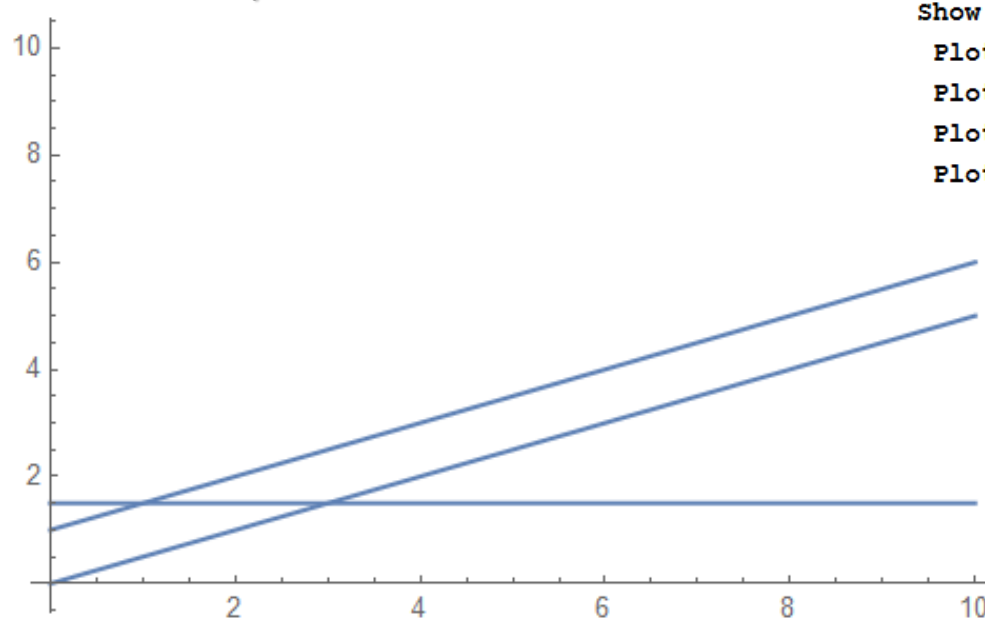


In this example, a model was built based on a linear regression with a variable
There are many different models, with different degrees of complexity





For the same problem we can create different models. How to choose the best?



```
Show[  
  Plot[1.5 + 0 * x, {x, 0, 10}],  
  Plot[0 + 0.5 * x, {x, 0, 10}],  
  Plot[1 + 0.5 * x, {x, 0, 10}],  
  PlotRange -> {{0, 10}, {0, 10}}, AxesOrigin -> {0, 0}]
```

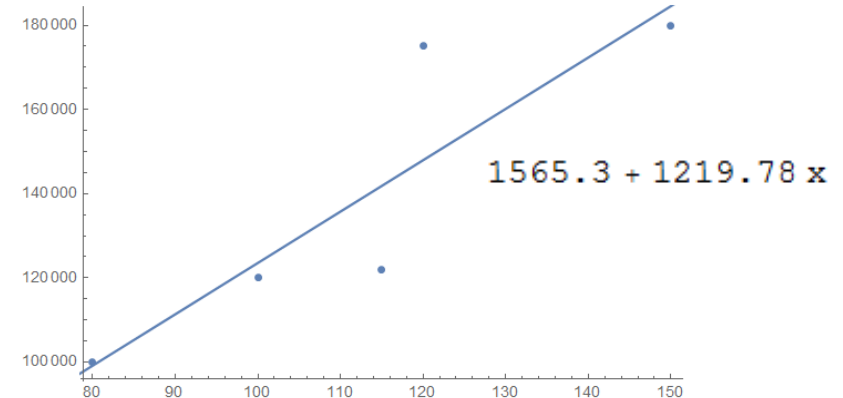
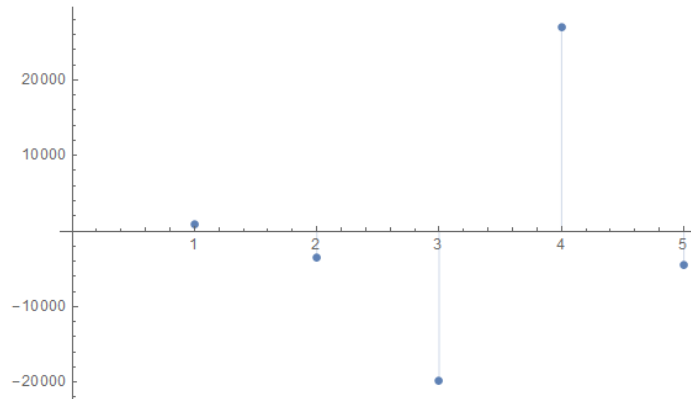
Key idea

Choose values of θ_0 and θ_1 that minimize the distance between $h_{\theta}(x)$ and y for each pair of values (x, y) in the training model

$$\text{minimize } \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

```
Total[Abs /@ lm["FitResiduals"]]
```

```
55 828.4
```



- **Supervised**

- Allow to train predictive models based on input and output data
- Allow to predict the outputs of new input values

- **Unsupervised**

- Group and interpret data based only on input data
- Allow pattern discovery

- **Reinforcement Learning**

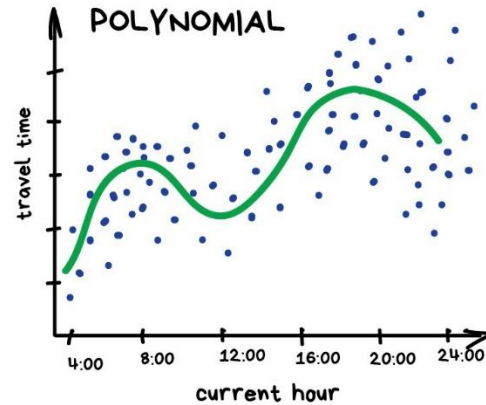
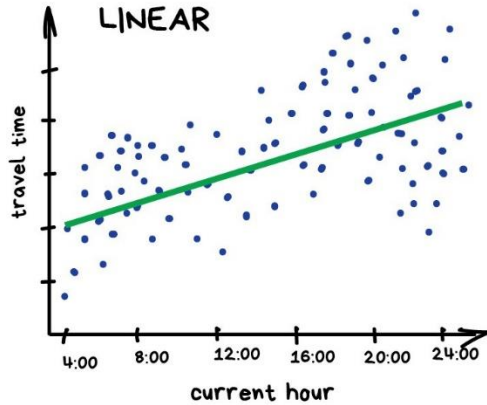
- No training data
- System learns by interacting with the environment and receiving rewards / penalties



ISLab

Synthetic Intelligence Lab

PREDICT TRAFFIC JAMS



Source: Machine Learning for Everyone
https://vas3k.com/blog/machine_learning/

REGRESSION

Supervised Learning Regression

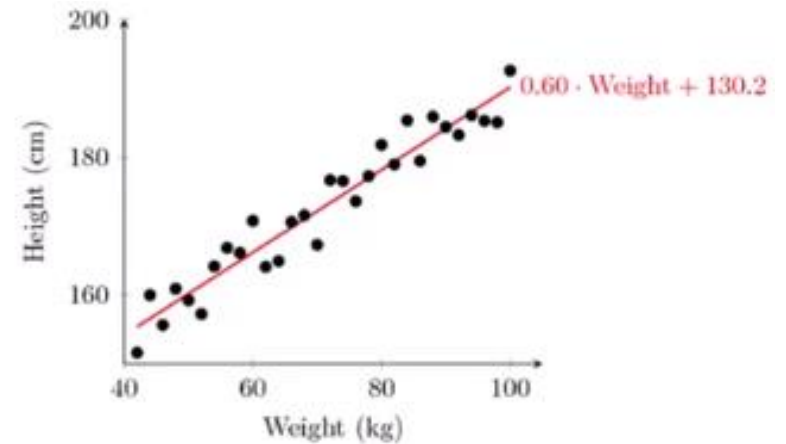


Source: Image: <https://www.flickr.com/photos/gsfrc/5937599688/>

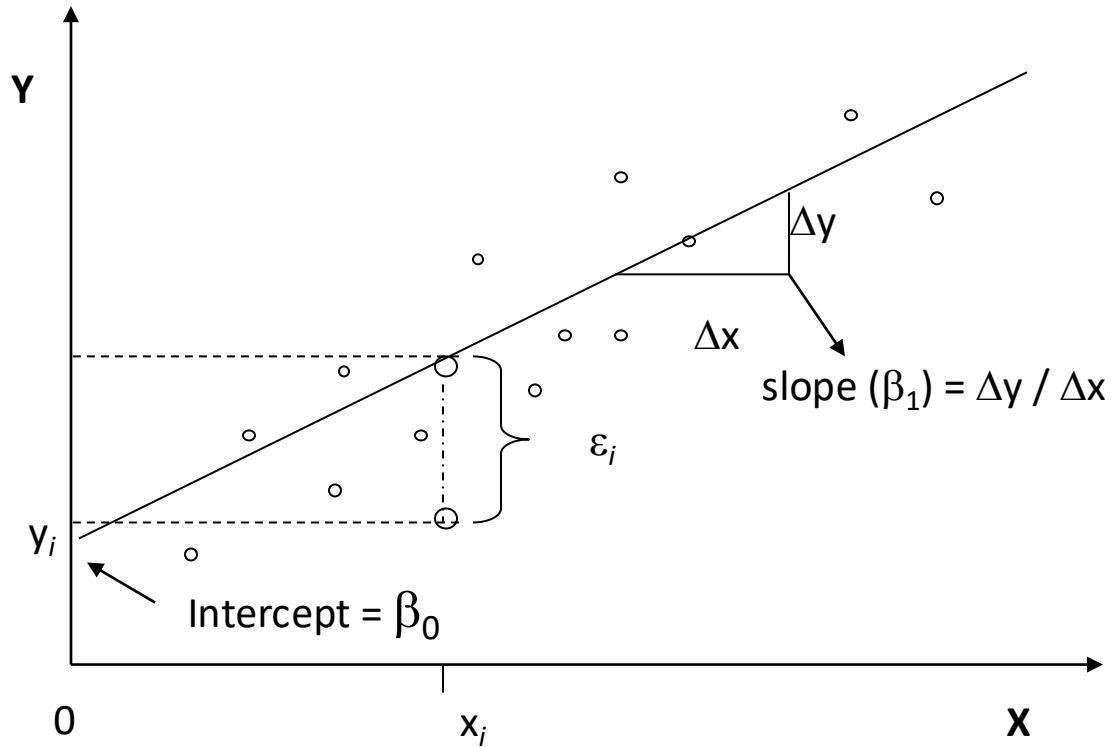
Linear Regression

Aims to predict the value of a outcome, Y, based on the value of an predictor variable, X.

- Fit a straight line into a data set of observations;
- Use this line to predict unobserved values.



Linear Regression

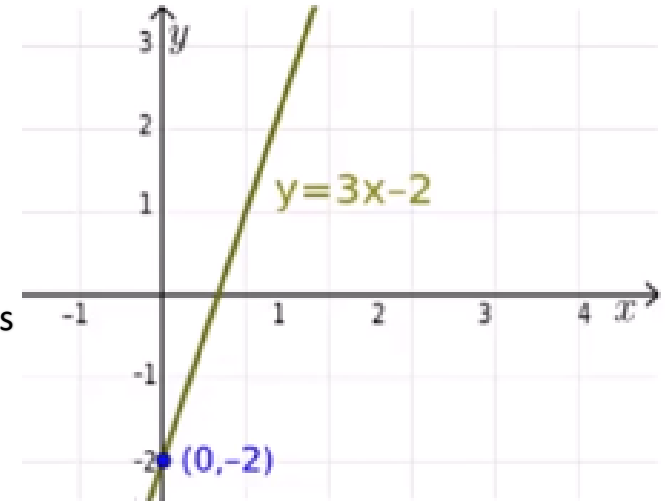


$$\text{Model: } Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

Linear Regression

How does it work?

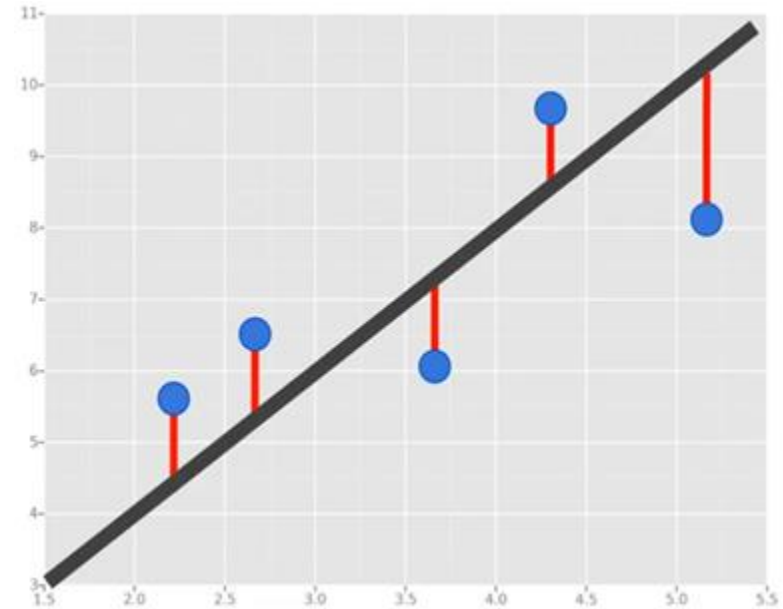
- Usually using “least squares” - minimize the squared-error between each point and the line
- Follows the slope-intercept equation of a line: $Y = m \cdot x + b$
 - m – slope: correlation between the two variables times the stand. Dev. In Y , all divided by the standard deviation in X .
 - b – y intercept: Intercept is the mean of Y minus the slope times the mean of X



Linear Regression

How does it work?

- Least squares minimizes the sum of squared errors
 - $y(i)$: true value
 - $F(x(i), \beta)$: predicted value / fitted line
- Residuals for an observation is the difference between the observation (y-value) and the fitted line



$$r_i = y_i - f(x_i, \beta).$$

The least-squares method finds the optimal parameter values by minimizing the sum, S , of squared residuals:

$$S = \sum_{i=1}^n r_i^2.$$

Measuring error with r-squared

- How do we measure how well our line fits our data?
- R-squared (i.e., coefficient of determination) measures – *“the fraction of the total variation in Y that is captured by the model”*

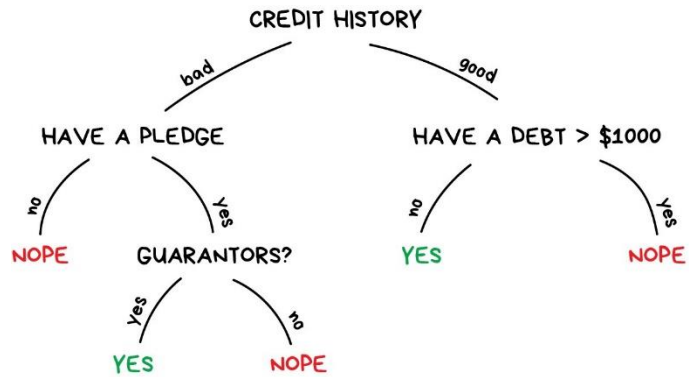
Computing r-squared

- Ranges from [0-1]
- 0 is bad (none of the variance is captured)
- 1 is good (all of the variance is captured)

$$1.0 - \frac{\text{sum of squared errors}}{\text{sum of squared variation from mean}}$$

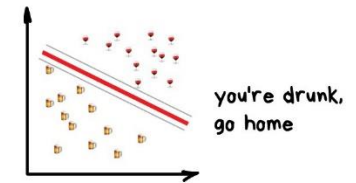
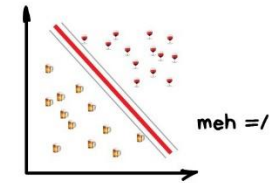
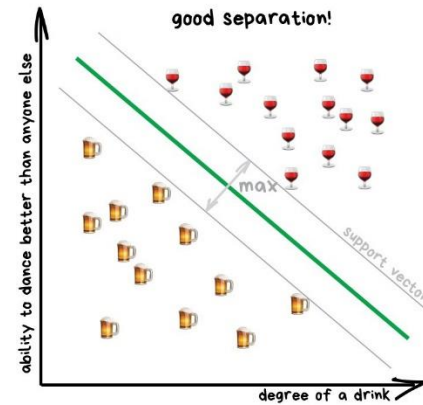
Supervised Learning Classification

GIVE A LOAN?



DECISION TREE

SEPARATE TYPES OF ALCOHOL



SUPPORT VECTOR MACHINE

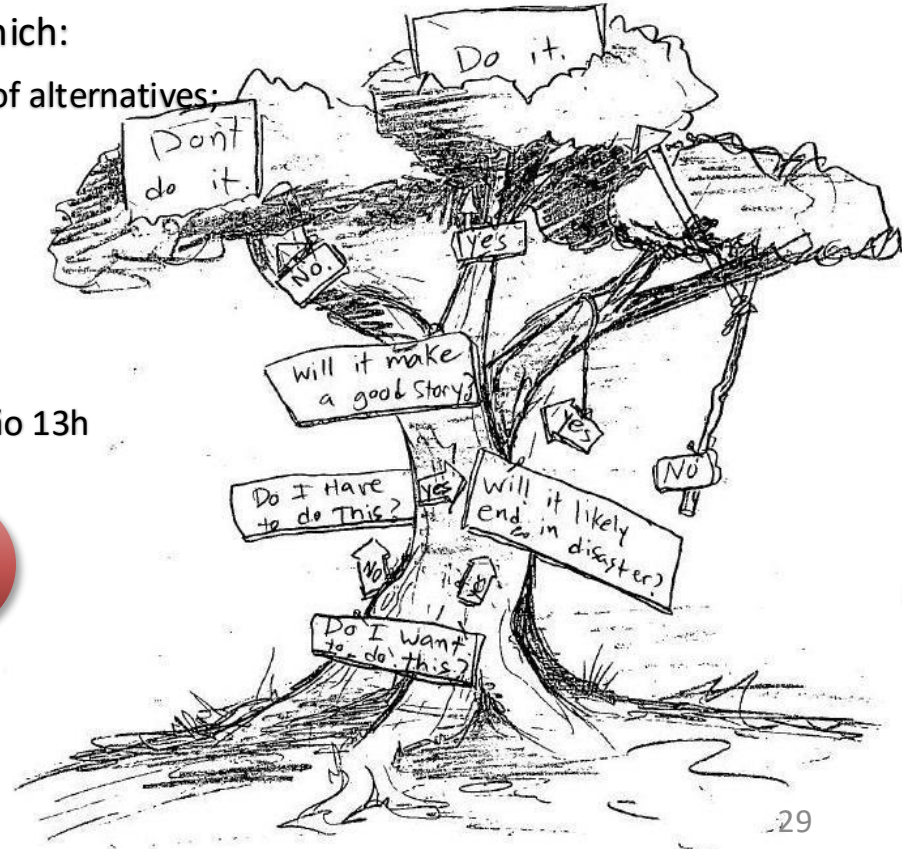
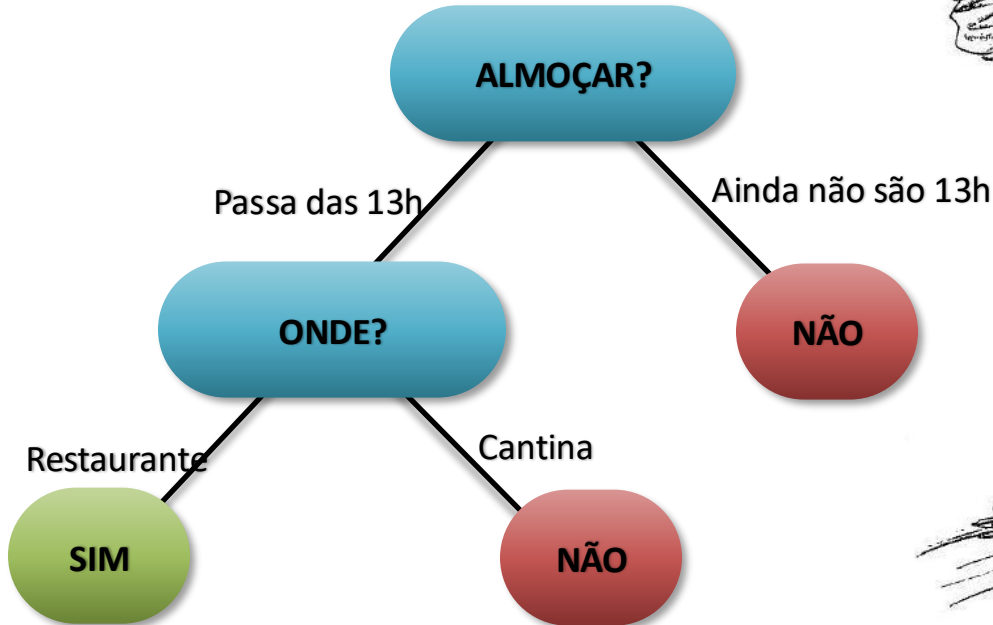


“Classification in machine learning is a supervised learning approach in which we learn from the data given to it and make new observations or classifications.”

- Given a collection of data (records) (training set)
- Each record is characterized by a tuple (x, y) , where x is the set of attributes and y is the assigned class or category.
 - x : attribute, predictor, independent variable, input
 - y : class, response, dependent variable, output
- Task:
 - Learn a model that maps each set of attributes x to one of y 's predefined classes

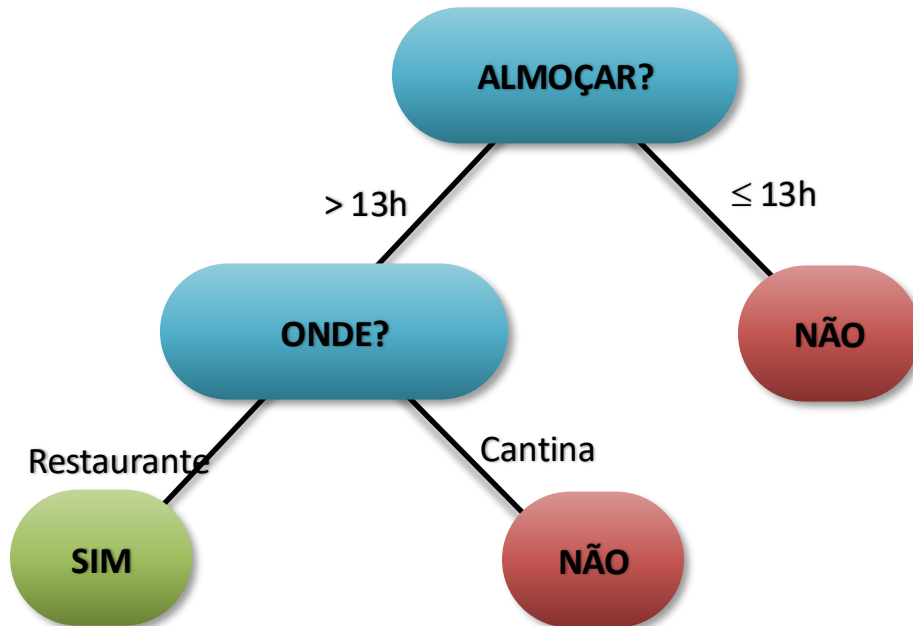
Decision Trees

- A Decision Tree is a hierarchical graph (tree!) in which:
 - Each branch represents the selection among a set of alternatives;
 - Each sheet represents a decision;



Decision Trees

- A Decision Tree is a hierarchical graph (tree!) in which:
 - Each internal node tests an attribute of the dataset;
 - Each branch identifies a value (or set of values) from the tested node;
 - Each sheet represents a decision;



ALMOÇOAR	ONDE	DECISÃO
12h30	Cantina	NÃO
13h15	Cantina	NÃO
13h10	Restaurante	SIM
11h00	Restaurante	NÃO
13:30	Cantina	NÃO

- Algorithm ID3: Iterative Dichotomiser 3

<https://www.cise.ufl.edu/~ddd/cap6635/Fall-97/Short-papers/2.htm>

- Developed by Ross Quinlan;
- Build a decision tree, from root to leaves;
- Main problem:
 - "What is the best attribute to be the root of the decision tree?"



Unsupervised Learning Clustering

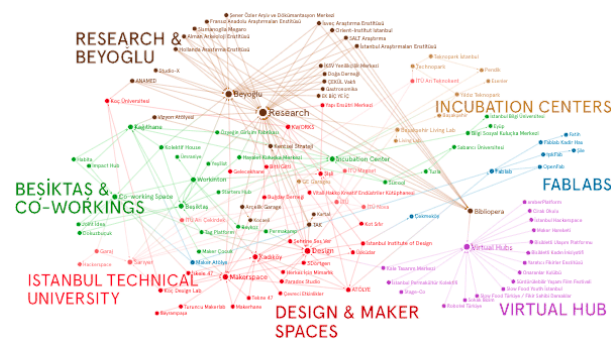


sample

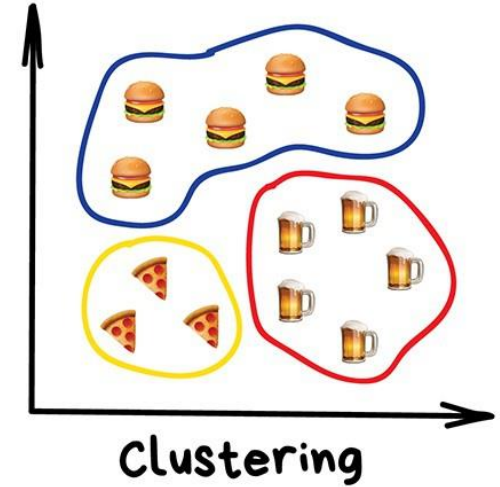


Cluster/group

Source: Unsupervised Machine Learning: What is, Algorithms, Example
<https://www.guru99.com/unsupervised-machine-learning.html>

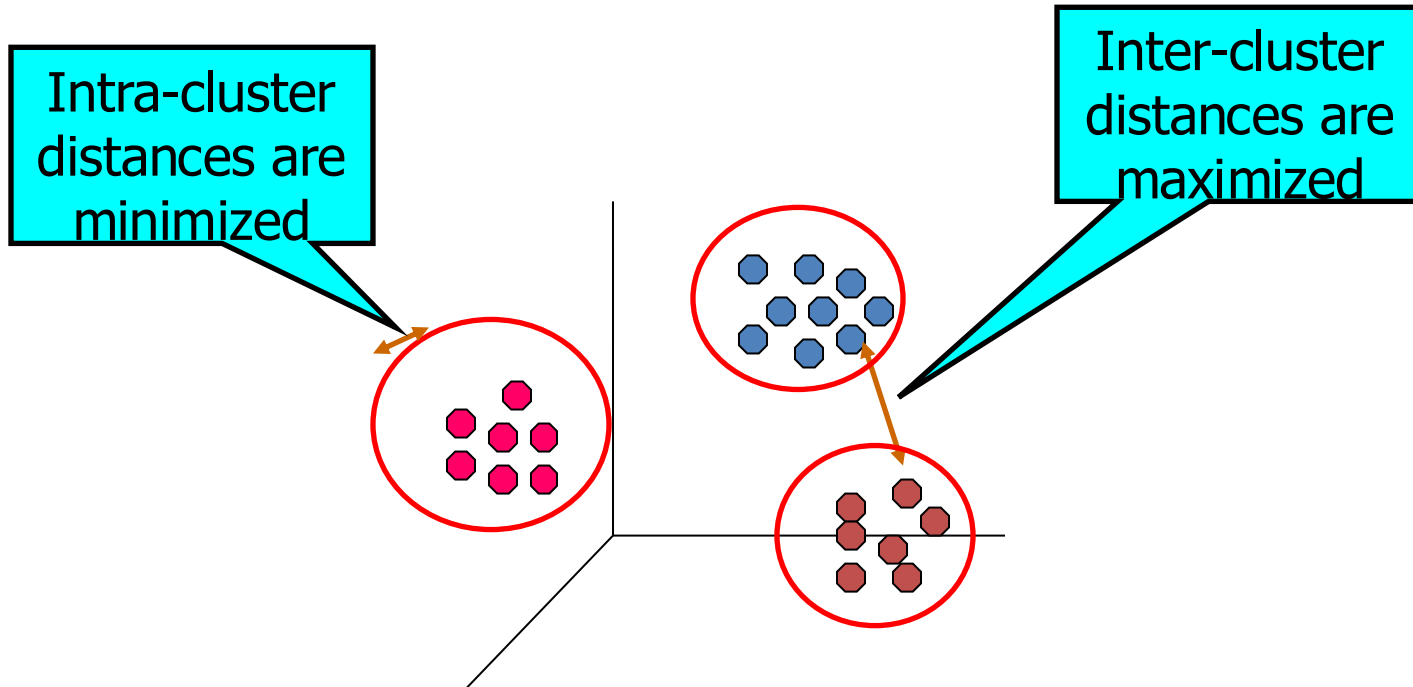


Source: <https://medium.com/graph-commons/finding-organic-clusters-in-your-complex-data-networks-5c27e1d4645d>



Source: Machine Learning for Everyone
https://vas3k.com/blog/machine_learning/

Given a set of objects, place them in groups such that the objects in a group are similar (or related) to one another and different from (or unrelated to) the objects in other groups

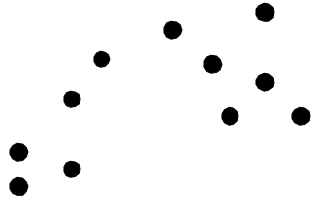




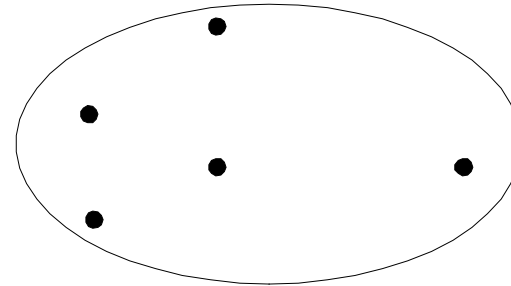
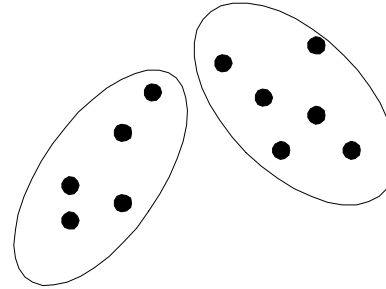
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Partitional Clustering

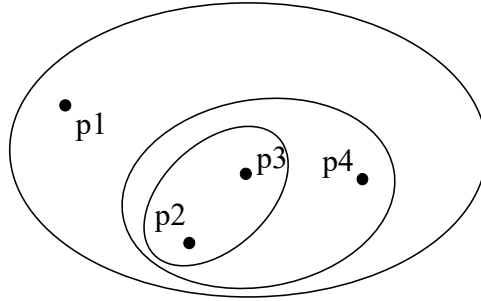


Original Points

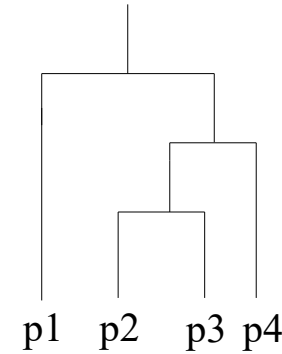


A Partitional Clustering

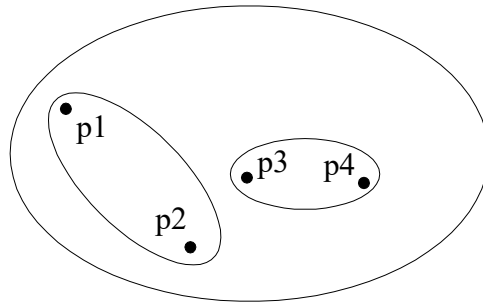
Hierarchical Clustering



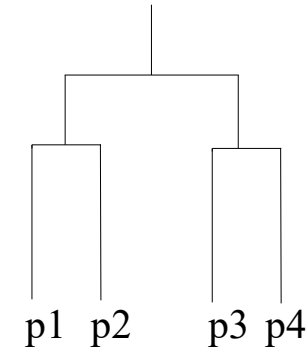
Traditional Hierarchical Clustering



Traditional Dendrogram



Non-traditional Hierarchical Clustering



Non-traditional Dendrogram

Textbook

- T. Michell, (2017) Machine Learning, McGraw Hill, ISBN ISBN 978-1259096952.
- E. Alpaydin, (2014) Introduction to Machine Learning. ISBN: 978-0-262-02818-9. Published by The MIT Press.
- Russell and Norvig, (2009) Artificial Intelligence - A Modern Approach, 3rd edition, ISBN-13: 978013604259.
- Pang Ning Tan, Michael Steinbach, Anuj Karpatne, Vipin Kumar (2018), Introduction to Data Mining, ISBN 9780133128901.

For those that want to go further

Pedro Domingos, (2015) The Master Algorithm: How the Will Remake Our World, Basic Books, ISBN: ISBN-13: 978

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